흉곽, 늑막삼출

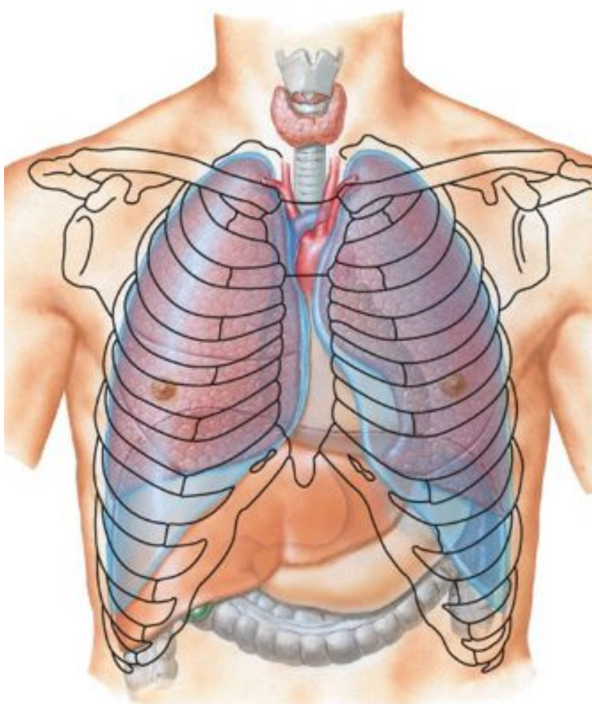
호흡기내과박 순 효

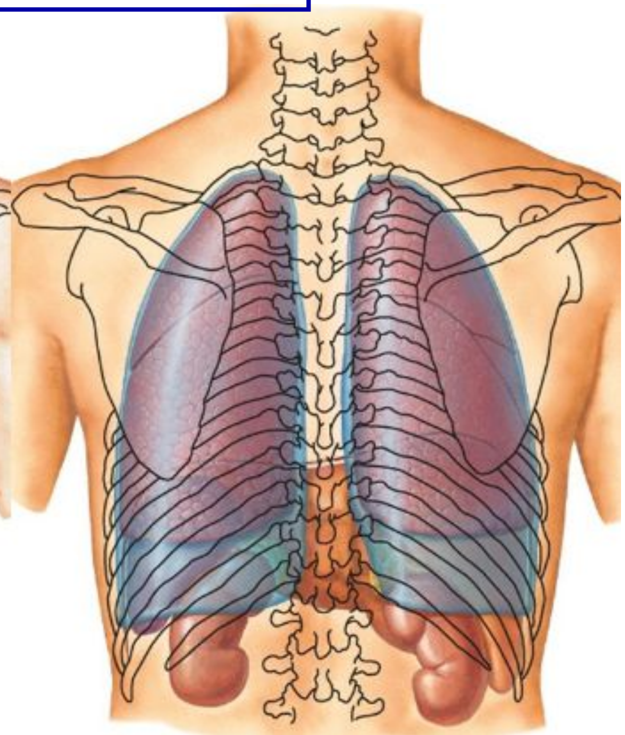
폐와 기관지의 위치와 구조

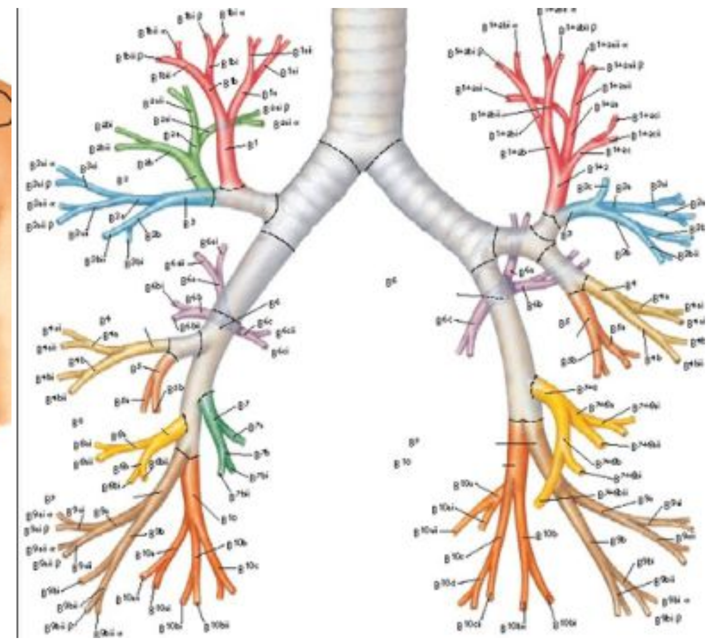
기관지-도로/폐-건물
폐와 기관지의 병변을 구분해서 부르지는 않는다

폐는 신경이 없다
늑막에는 신경이 있다→흉수가 찰 경우 통증이 매우 심할 수 있다(개인차가 있다)

한 군데에서만 Wheezing이 들린다면 암일 가능성이 가장 높다

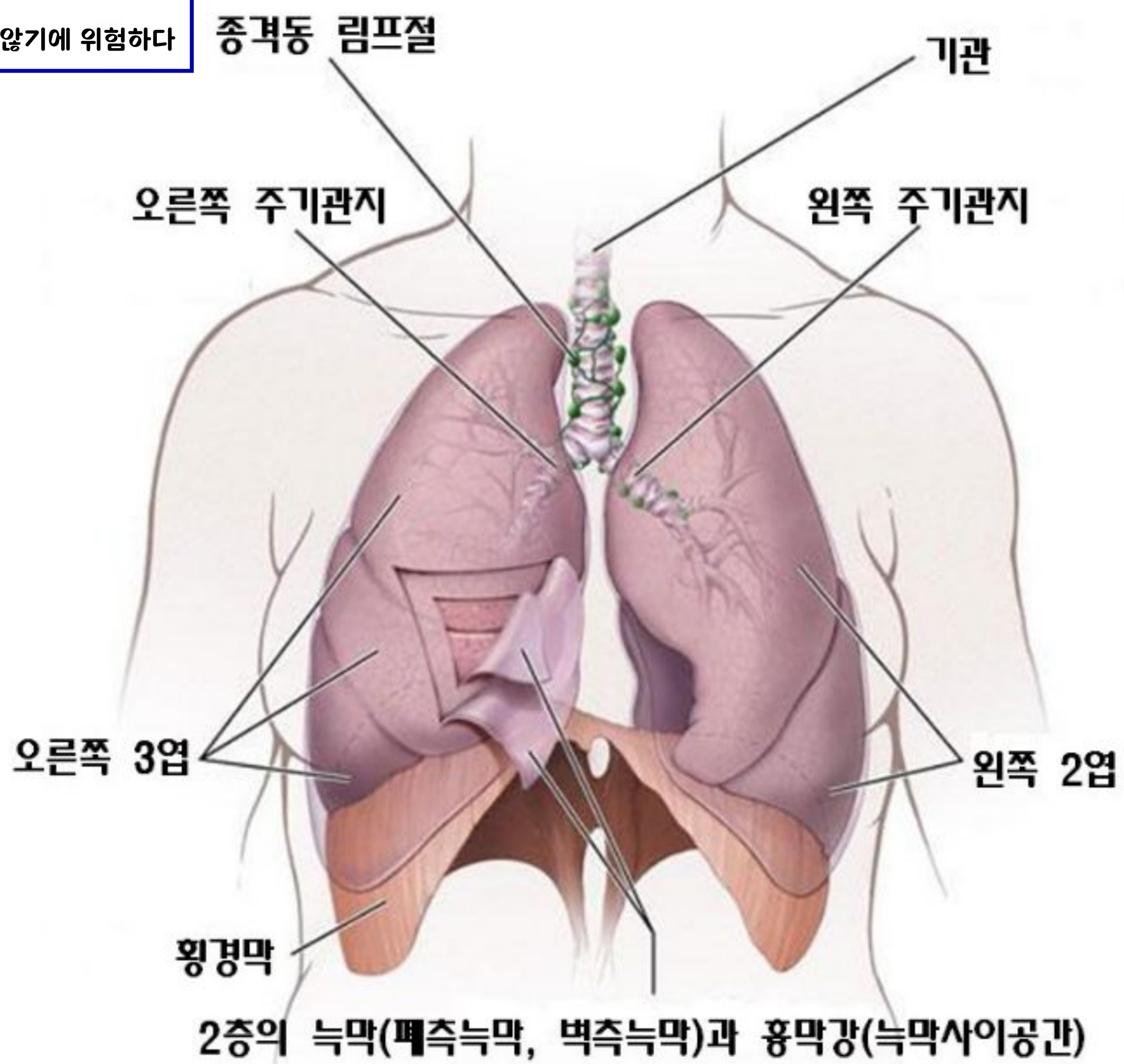






폐와 기관지의 위치와 구조

간암/췌장암/담낭암/폐암은
신경이 없어 통증이 느껴지지 않기에 위험하다



1. 흉수의 발생기전

- Pleural fluid formation exceeds pleural fluid absorption.
- 정상: small amount (0.01 ml/kg/hr) of fluid

Parietal/Visceral의 마찰력을 줄여주는 역할 (윤활유 역할)

점도가 높아질 경우 통증이 생길 수 있다

- Parietal pleura: capillaries

  interstitial space of lung-visceral pleura

  peritoneal cavity-small holes in the diaphragm

  → pleural space

  → parietal pleura: lymphatics
- The lymphatics have the capacity to absorb 20 times more fluid than is formed normally.

폐측

Cancer가 생기면 lymphatic drainage 장애
→물이 참

결핵이 생기면 물 생성 자체가 많아짐
→물이 참

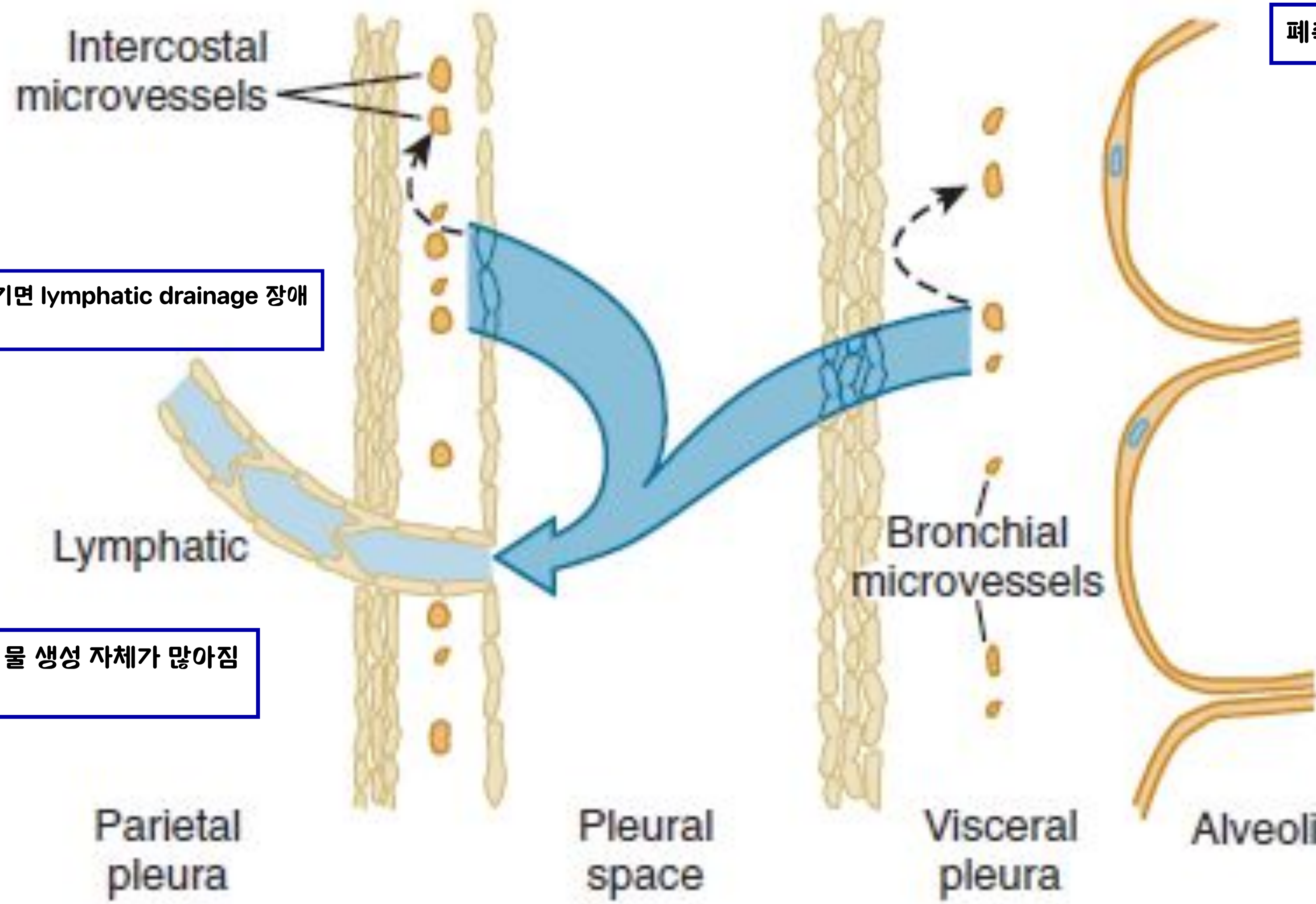


Figure 79-3 Schema showing normal pleural liquid turnover. The

Pleural effusion may develop when there is excess pleural fluid formation (from the interstitial spaces of the lung, the parietal pleura, or the peritoneal cavity) or when there is decreased fluid removal by the lymphatics.

Increased pleural effusion

- Increased interstitial fluid
- Increased hydrostatic pressure gradient Vessel→Pleural space로 미는힘
- Increased capillary permeability
- Decreased oncotic pressure gradient Pleural space가 vessel보다 삼투압이 높아야함
- Presence of free peritoneal fluid or disruption of the thoracic duct Cancer cell에 의해 막힘
- Decreased pleural fluid absorption

2. 진단적 접근법

1) 임상증상, 신체검사

늑막통

: dyspnea, pleuritic chest pain

늑막통은 숨을 쉬며 Parietal pleura와 Visceral pleura가 만나며 발생

물이 너무 많으면
→통증을 느끼지 못한다(∵Parietal,Visceral pleura가 만나지 못하기에)
&Dyspnea(∵폐가 눌려 Tidal volume 감소)

: decreased breath sounds

dullness to percussion

egophony at the upper level of the effusion

decreased tactile fremitus

2) Chest imaging

① chest X-ray

→ lateral decubitus view 매우 중요하다
(하지만 액체의 성분을 점도 정도만 추정할 뿐,
정확히 알기 위해서는 천자가 필요하다)

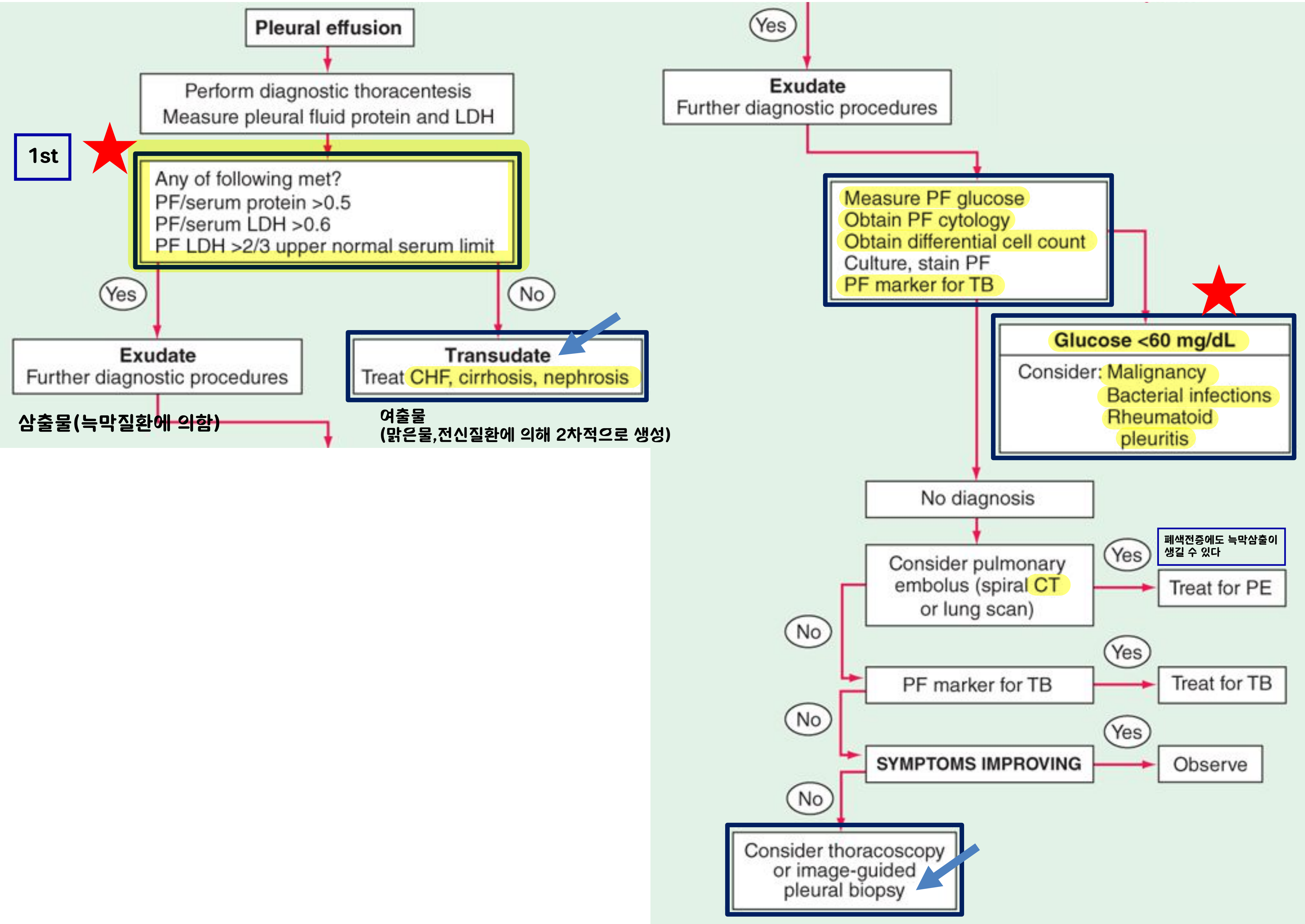
② chest ultrasound 천자를 곧바로 시행할 수 있기에 초음파의 활용도가 더 크다

Chest ultrasound has replaced the lateral decubitus x-ray in the evaluation of suspected pleural effusions and as a guide to thoracentesis.

처음에 한장은 양쪽으로 찍어야 함

반드시 알고리즘의 순서를 따라야함

Approach to the diagnosis of pleural effusions



3) Diagnostic thoracentesis

여출액 transudate vs 삼출액 exudate

- Exudate criteria(Light's critiera) : at least one

① pleural fluid/ serum protein > 0.5

② pleural fluid/ serum LDH > 0.6

③ pleural fluid LDH

> 2/3 upper normal serum limit (200)

* misidentify- 25% of transudate as exudate

* transudate가 의심되면,

serum-pleural fluid protein gradient >3.1g/dl

예외가 있을 수 있다고 생각은 해야 함
(Gold standard는 아니다)

3. 흉수의 원인

- Transudate vs Exudate

Differential Diagnosis of Pleural Effusion

Transudative Pleural Effusions

1. Congestive heart failure
2. Cirrhosis
3. Nephrotic syndrome
4. Peritoneal dialysis
5. Superior vena cava obstruction
6. Myxedema
7. Urinothorax

Exudative Pleural Effusions

1. Neoplastic diseases
   a. Metastatic disease
   b. Mesothelioma
2. Infectious diseases
   a. Bacterial infections
   b. Tuberculosis
   c. Fungal infections
   d. Viral infections
   e. Parasitic infections
3. Pulmonary embolization
4. Gastrointestinal disease
   a. Esophageal perforation
   b. Pancreatic disease
   c. Intraabdominal abscesses
   d. Diaphragmatic hernia
   e. After abdominal surgery
   f. Endoscopic variceal sclerotherapy
   g. After liver transplant
5. Collagen vascular diseases
   a. Rheumatoid pleuritis
   b. Systemic lupus erythematosus
   c. Drug-induced lupus
   d. Sjögren syndrome
   e. Granulomatosis with polyangiitis (Wegener)
   f. Churg-Strauss syndrome

Exudative Pleural Effusions

6. Post–coronary artery bypass surgery
7. Asbestos exposure
8. Sarcoidosis
9. Uremia
10. Meigs' syndrome
11. Yellow nail syndrome
12. Drug-induced pleural disease
    a. Nitrofurantoin
    b. Dantrolene
    c. Methysergide
    d. Bromocriptine
    e. Procarbazine
    f. Amiodarone
    g. Dasatinib
13. Trapped lung
14. Radiation therapy
15. Post–cardiac injury syndrome
16. Hemothorax
17. Iatrogenic injury
18. Ovarian hyperstimulation syndrome
19. Pericardial disease
20. Chylothorax

부폐렴성 흉수

빈출주제

Parapneumonic effusion

1. 정의

: associated with bacterial pneumonia,
lung abscess, bronchiectasis

: M/C cause of exudative pleural effusion in US

* 농흉 (empyema): grossly purulent effusion

폐렴 동반할 수도/안할 수도 있다

2. 증상

m/c

1) aerobic bacteria: acute febrile illness

- chest pain, sputum production, leukocytosis

2) anaerobic bacteria: subacute illness

- weight loss, brisk leukocytosis, mild anemia, aspiration history

3. 진단

lateral decubitus view: free fluid shifting >10mm

→ thoracentesis

4. 치료

1) Antibiotics Tazocin(Tazobactam+Piperacillin)

2) Chest tube insertion 배액이 치료에 중요한 역할

가슴관삽입
적응증

① loculated pleura effusion 흐르지 않는 경우

② pH < 7.20

③ PF glucose < 60mg/dl

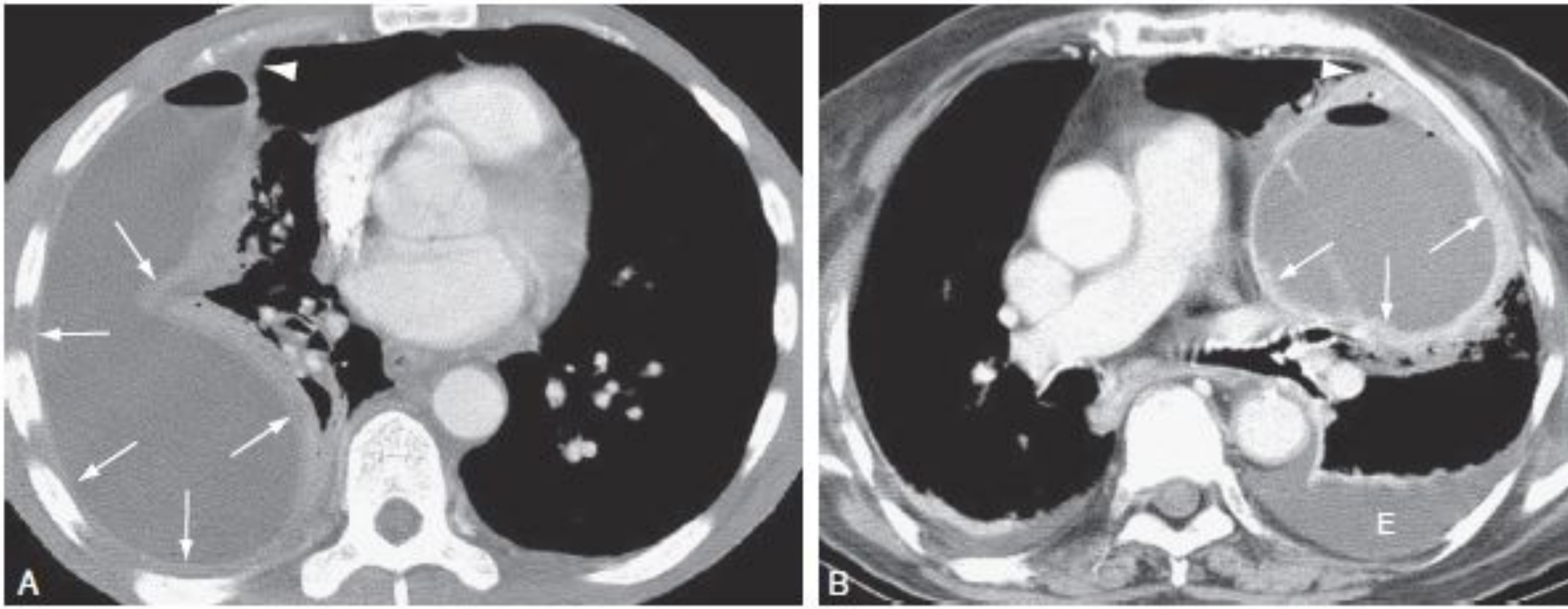
④ Gram stain or culture (+) of PF

⑤ gross pus in Pleural space

3) Instillation of Fibrinolytic agents (ex.tPA)

4) Thoracoscopy with the breakdown of adhesion

5) Decortication Thoracoscopy통해 폐기능 유지를 위해 치료


A
B
E

결핵성 흉막염
Tuberculous pleurisy

1. 원인

: MC cause of an exudate pleural effusion

: associated with primary TB

: hypersensitivity reaction to tuberculous protein in the pleural space

균의 수가 많아서가 아니라, 과민반응에 의해 흉수의 양이 많다
(물의 양이 중증도를 결정짓지 않는다)

2. 증상

(Mild)
: fever, weight loss, dyspnea, and/or pleuritic chest pain

3. 진단

1st)Exudate/Transudate
2nd)Glucose 체크
3rd)ADA>40
(알고리즘의 순서를 잘 따라가며 진단해야 함)

1) exudate with predominantly lymphocytes

2) high level of TB markers in the pleural fluid

: ADA (adenosine deaminase) > 40 IU/L

- Empyema, Rheumatoid arthritis, Lymphoma 증가할 수.

: Interferon-gamma >140 pg/ml

민감도는 더 높으나, 가격이 비싸 상용화되지 않음

: MTB-PCR (+)

3) PF culture(+), pleural biopsy, thoracoscopy

4. 치료: anti-TB medication

여출액
Transudative pleural effusion

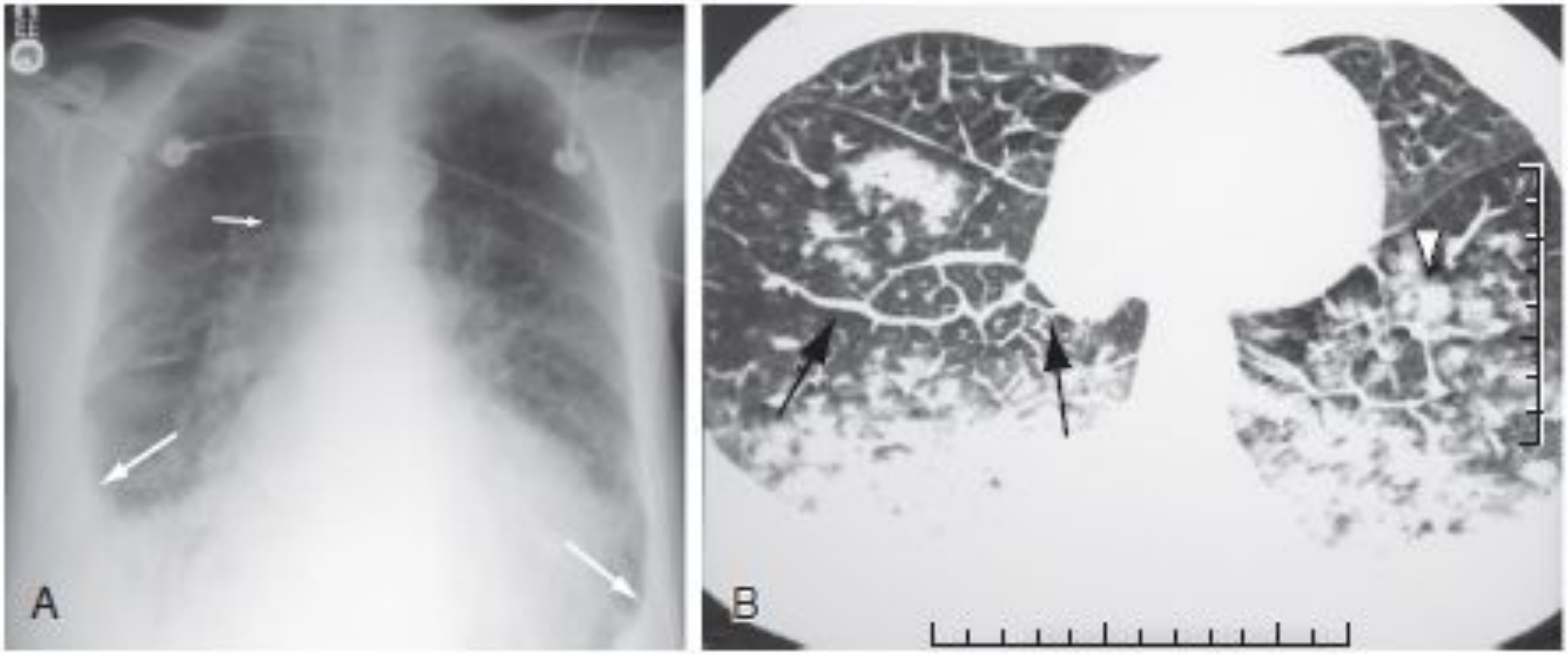
1. Heart failure

- Most common cause of pleural effusion is left heart failure
- Increased amounts of fluid in the lung interstitial spaces
- Diagnostic thoracentesis

  : not bilateral & comparable size, fever, pleuritic chest pain, effusion persist despite therapy
- Pleual fluid NT-proBNP > 1500 pg/ml

Interseptal thickening이 생긴다

대부분 heart가 크다



2. Hepatic hydrothorax

- 5% of cirrhosis & ascites
- Peritoneal fluid → diaphragm: small openings → pleural space
- Usually right-sided
- frequently is large enough to produce severe dyspnea

악성 흉수

Malignant pleural effusion

Malignant pleural effusion

- 2nd M/C type of exudative pleural effusion

Stage 4이상
→항암치료 해야함

- Lung cancer, breast cancer, lymphoma
  - 3 tumors 75% of all malignant pleural effusion
  - others (ovarian cancer, gastric cancer)
- Sx.
  - dyspnea(out of portion to the size of the effusion)
- Exudate (or rarely transudate)
  - glucose level may be reduced.↓(if tumor burden in the pleural space ↑)
- Benign effusions in patient with underlying malignancy
  - atelectasis, postobstructive pneumonia, hypoalbuminemia, pulmonary emboli, lymphatic obstruction, complications from radiation, chemotherapy

Malignant pleural effusion

- Diagnosis
  - cytology of the pleural fluid (malignant cells in pleural effsion)
  - 60% can be diagnosed with one thoracentesis
  - initial cytology is negative

    → thoracoscopy

    → CT- or ultrasound-guided needle biopsy
- Treatment
  - symptomatic treatment
    - d/t not curable with chemotherapy
  - therapeutic thoracentesis
    - insertion of a small indwelling catheter
    - tube thoracostomy with installation of a sclerosing agent (doxycycline, talc)

Management of Malignant and Paramalignant Pleural Effusions

Option	Comment
Observation	Small asymptomatic effusion; most will progress and require therapy
Therapeutic thoracentesis	Prompt relief of dyspnea; recurrence rate variable
Chemotherapy	May be effective in lymphoma, small-cell lung cancer, breast cancer
Radiotherapy	Mediastinal radiation may be effective in lymphoma and lymphomatous chylothorax
Indwelling catheter	Patient controlled symptom relief; spontaneous pleurodesis in 50% by 2 months. Effective for symptomatic relief with lung entrapment.
Chest tube drainage with talc slurry	Control of effusion in >90 percent of cases if lung entrapment not present
Thoracoscopy with talc poudrage	Control of effusion in >90 percent of cases if lung entrapment not present
Pleuroperitoneal shunt	When other options have failed or not indicated; may be useful for chylothorax
Pleural abrasion and partial pleurectomy	Virtually 100 percent effective; requires VATS or thoracotomy

중피종
Mesothelioma

Malignant mesothelioma

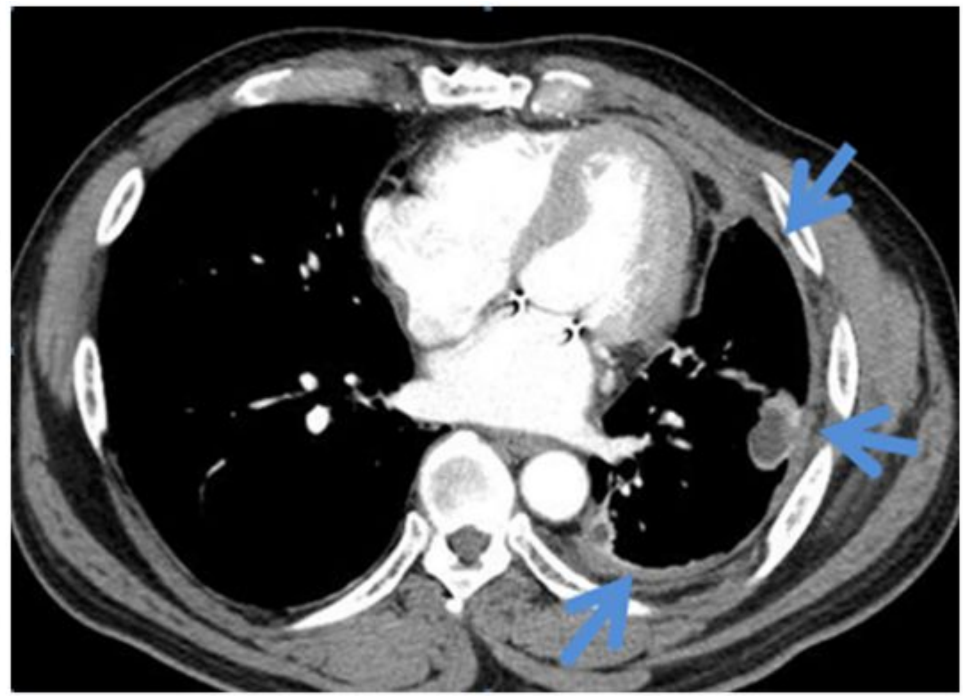
- Primary tumors that arise from the mesothelial cells that line the pleural cavities
- most common primary malignant tumor of pleura
- most related to asbestos exposure(70%)
  - 1-2년의 짧은 노출에도 유발가능
  - 노출 30-35년 뒤가 peak

직업성 질환과 연관지어 출제 가능성

- Sx. : chest pain, shortness of breath
- CXR : pleural effusion, generalized pleural thickening, shrunken hemithorax, pleural plaques(be calcified)
- Dx. : image-guided needle biopsy or thoracoscopy

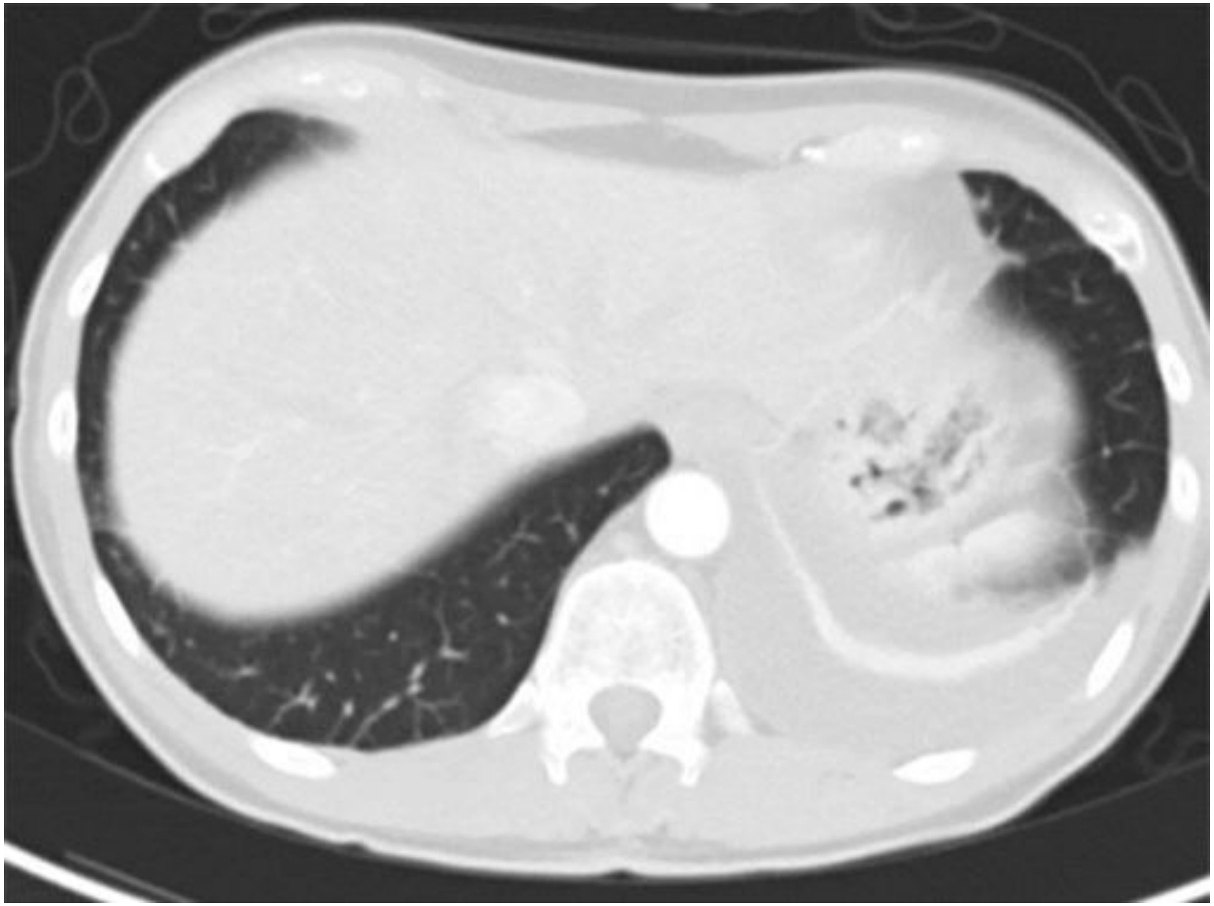
Pleural biopsy를 통해서만 진단이 가능함

Malignant mesothelioma



Pleural plaque

Malignant mesothelioma



CT scan from a patient with mesothelioma demonstrating a mass in the left lung, a pleural effusion, pleural thickening, and a shrunken hemithorax.

폐색전증
Pulmonary Embolism

Pulmonary Embolism

- MC overlooked in the differential diagnosis of undiagnosed pleural effusion
- M/C Symptom : dyspnea
- Almost always Exudate
- Diagnosis : CT scan or pul. Arteriography
- Treatment : same as PE.
  - If the pleural effusion increases after anticoagulation
    → recurrent emboli, hemothorax or pleural infection.

Pulmonary infartion에 의해 생김
(Secondary이기에 Transudate라고 생각할 수 있지만, Exudate이다)
(Transudate는 심장/간/신장이 원인)

바이러스감염
Viral Infection

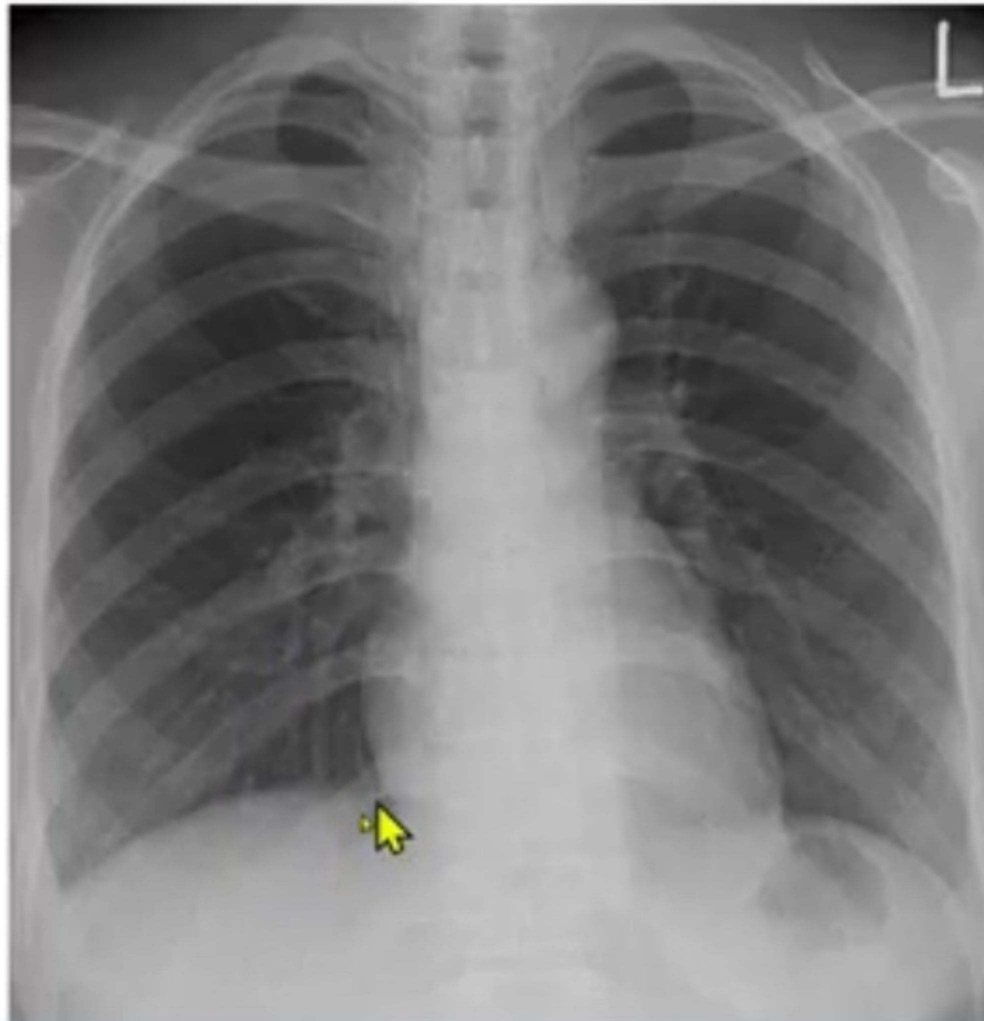
Viral Infection

- Sizable percentage of undiagnosed exudative pleural effusions 자연회복되는 경우 多
- no diagnosis is established for 20% of exudative effusions → resolve spontaneously with no long-term residua
- The importance of these effusions is that one should not be too aggressive in trying to establish a diagnosis for the undiagnosed effusion, particularly if the patient is improving clinically.

Blunting이 잘 보이지 않음

F/41

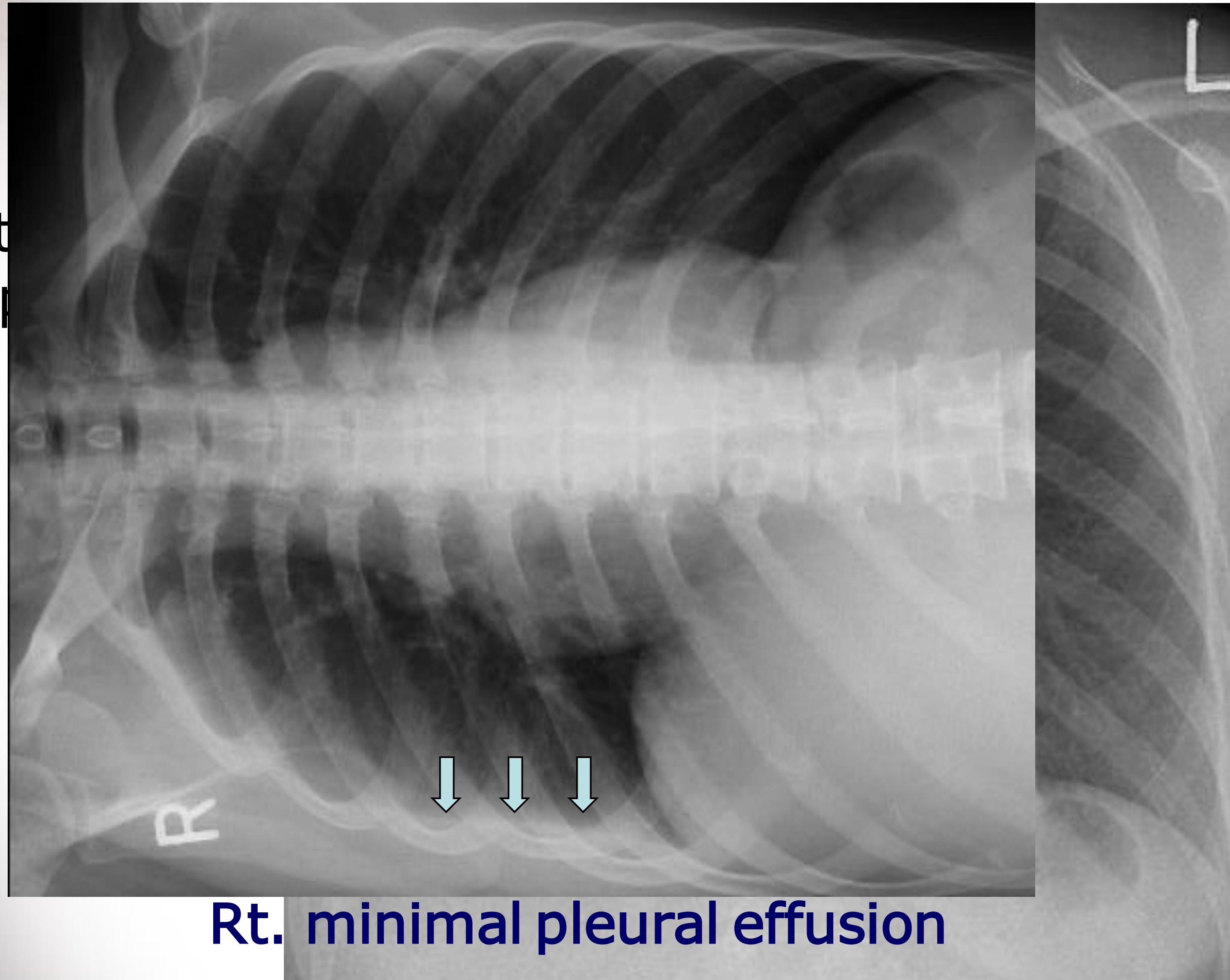
C/C Rt. pleuritic chest pain



옆으로 눕히면 Effusion이 보임

F/41

C/C Rt
chest p

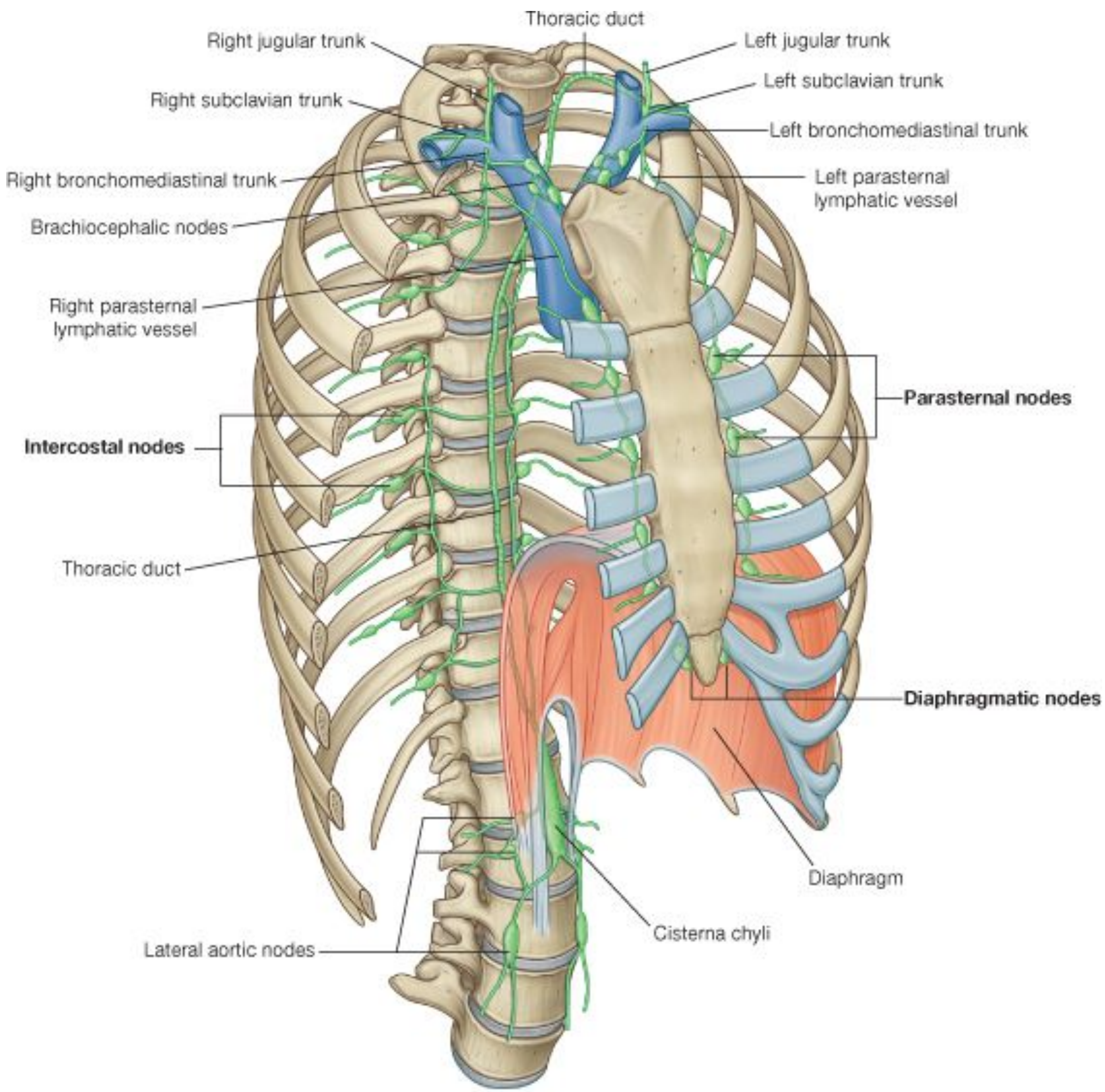


Rt. minimal pleural effusion

유미흉
Chylothorax

Chylothorax

- Chyle :
  - milky bodily fluid consisting of lymph and emulsified fats, or free fatty acid.
  - formed in the small intestine during digestion of fatty foods
- Chylothorax : thoracic duct is disrupted and chyle accumulates in the pleural space
- Cause : Most frequently trauma(thoracic surgery), tumor in mediastimum
- Pleural fluid : milky, TG > 110mg/dL cholesterol crystal 없다.
  - 50< TG <110 ; lipoprotein analysis -> chylomicron 확인
  - TG < 50 ; pseudochylothorax, cholesterol >250, cholesterol crystal 관찰, 결핵성 만성농흉이 가장 흔하고, 그 외 RA에서 가능.
  - If no obvious trauma → Lymphangiogram & mediastinal CT scan
- Treatment
  - insertion of a chest tube plus the administration of octreotide
  - percutaneous transabdominal thoracic duct blockage
  - ligation of the thoracic duct
  - should not undergo prolonged tube thoracostomy with chest tube drainage because this will lead to malnutrition and immunologic incompetence


Thoracic duct
Right jugular trunk
Left jugular trunk
Left subclavian trunk
Right subclavian trunk
Left bronchomediastinal trunk
Right bronchomediastinal trunk
Left parasternal lymphatic vessel
Brachiocephalic nodes
Right parasternal lymphatic vessel
Parasternal nodes
Intercostal nodes
Thoracic duct
Diaphragmatic nodes
Diaphragm
Cisterna chyli
Lateral aortic nodes

Chylothorax

혈흉
Hemothorax

Hemothorax

- Significant amount of blood in the pleural space
- Cause
  - M/C cause : result of trauma
  - rupture of a blood vessel or tumor
- Bleeding site
  - MC : pulmonary parenchymal vessel(low pressure)
  - common : pulmonary parenchymal laceration, intercostal vessel laceration, rupture of plueral adhesions
  - less common : mediastinal injury (damage of a major blood vessel or decompression of abdominal hemorrhage)
- 60-80% of hemothotrax patients is associated pneumothorax.
- Dignosis
  - bloody PF → hematocrit should be obtained on the pleural fluid
  - PF Hct >50% of peripheral blood Hct → Consider hemothorax

Hemothorax

- Bloody pleural effusion causes
  - malignancy, trauma, pulmonary infarction, tuberculosis, hematologic disorder

- Treatment
  - TOC ; Tube thoracostomy (control 85% bleeding)
  - If the bleeding emanates from a laceration of the pleura
  → apposition of the two pleural surfaces is likely to stop bleeding

  - If Pleural hemorrhage exceeds 200mL/h
  →Consider angiographic coil embolization, thoracoscopy, or thoracotomy

수술적 치료 고려

Differential Diagnosis of Pleural Effusion

Transudative Pleural Effusions
1. Congestive heart failure
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3. Nephrotic syndrome
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13. Trapped lung
14. Radiation therapy
15. Post–cardiac injury syndrome
16. Hemothorax
17. Iatrogenic injury
18. Ovarian hyperstimulation syndrome
19. Pericardial disease
20. Chylothorax

Correlation of Pleural Fluid Exudate Findings & Causative Disease

Tests	Diseases
pH <7.2	Empyema, malignancy, esophageal rupture; rheumatoid, lupus, and tuberculous pleuritis
Glucose (<60 mg/dL)	Infection, rheumatoid pleurisy, tuberculous and lupus effusions, esophageal rupture
Amylase (>200 μg/dL)	Pancreatic disease, esophageal rupture, malignancy, ruptured ectopic pregnancy
Rheumatoid factor, antinuclear antibody, LE cells	Collagen vascular disease
Complement (decreased)	Lupus erythematosus, rheumatoid arthritis
Red blood cells (>5000/μL)	Trauma, malignancy, pulmonary embolus
Chylous effusion (triglycerides >110 mg/dL)	Tuberculosis, violation of the thoracic duct (trauma, malignancy)
Biopsy (+)	Malignancy
Adenosine deaminase (>40 μg/L)	Tuberculosis

LE = lupus erythematosus.

Exudate 소견과 원인질환

TEST	DISEASE
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Glucose (<60 mg/dL)	Infection, rheumatoid pleurisy, tuberculous and lupus effusions, esophageal rupture
Amylase (>200 μg/dL)	Pancreatic disease, esophageal rupture, malignancy, ruptured ectopic pregnancy
RF, ANA, LE cells	Collagen vascular disease
↓ Complement	SLE, RA
RBCs (>5000/μL)	Trauma, malignancy, pulmonary embolus
Chylous effusion (triglycerides > 110 mg/dL)	Tuberculosis, disruption of thoracic duct (trauma, malignancy)
Cytology or biopsy (+)	Malignancy
ADA (>50 μg/L)	Tuberculosis

ADA = adenosine deaminase; ANA = antinuclear antibody; RA = rheumatoid arthritis; RBC = red blood cell; RF, rheumatoid factor; SLE = systemic lupus erythematosus.

Miscellaneous Causes of Pleural Effusion

There are many other causes of pleural effusion.

Key features of some of these conditions are as follows:

국시에 나왔던 주제

If the pleural fluid amylase level is elevated, the diagnosis of esophageal rupture or pancreatic disease is likely.

If the patient is febrile, has predominantly polymorphonuclear cells in the pleural fluid, and has no pulmonary parenchymal abnormalities, an intra-abdominal abscess should be considered.

The diagnosis of an asbestos pleural effusion is one of exclusion.

Benign ovarian tumors can produce ascites and a pleural effusion (Meigs' syndrome), as can the ovarian hyper-stimulation syndrome.

Several drugs can cause pleural effusion; the associated fluid may be eosinophilic.

Pleural effusions commonly occur after coronary artery bypass graft (CABG) surgery. Effusions occurring within the first weeks after CABG are typically left-sided and bloody, with large numbers of eosinophils, and respond to one or two therapeutic thoracenteses. Effusions occurring after the first few weeks of CABG are typically left-sided and clear yellow, with lymphocytic predominance, and tend to recur.

Other medical manipulations that induce pleural effusions include abdominal surgery; radiation therapy; liver, lung, or heart transplantation; and complications of the intravascular insertion of central lines.

Fibrothorax (섬유흉)

- Results from fibrosis of the visceral pleura surrounding the lung
- decreased respiratory excursion and restrictive pulmonary physiology
- causes
  - pleural inflammation in patients with pleural effusion(common)
    - incompletely evacuated hemothorax, tuberculous effusion, chronic empyema
  - paranchymal disease(less common)
    - inadequately treated tuberculosis, bronchiectasis, lung abscess
- Treatment
  - observation
  - surgical decortication
    - limited indication; according to patient's sx. , lung function

맥박,호흡,체온 증가→세균 감염

63세 남자가 3일 전부터 왼쪽 가슴이 아파 병원에 왔다. 2주일 전부터 기침, 가래와 함께 발열, 오한이 있었다. 3일 전부터는 숨을 들이쉴 때 왼쪽 가슴에 뜨끔하는 통증이 생겼다. 혈압 128/86 mmHg, 맥박 101회/분, 호흡 22회/분, 체온 38.3℃였다. 가슴 청진에서 왼쪽 호흡음이 감소되었다. 혈액과 가슴막삼출액 검사 결과는 다음과 같았다. 세프트리악손(ceftriaxone)과 클린다마이신(clindamycin) 으로 치료를 시작하였다. 추가 치료는?

혈액: 혈색소 13.2 g/dL, 백혈구 11,300/mm3 혈소판 350,000/mm3,
단백질 6.6 g/dL, 포도당 102 mg/dL 젖산탈수소효소 430 U/L

가슴막삼출액: pH 6.7, 백혈구 12,100/mm3 (중성구 81%, 림프구 19%)
단백질 4.1 g/dL, 포도당 25 mg/dL 젖산탈수소효소 1,100 U/L
아데노신탈아미노효소(ADA) 96 IU/L 암배아항원 1.4 ng/mL

① 항결핵제
② 가슴관삽입
③ 가슴막유착술
④ 겉질제거(decortication)
⑤ 반코마이신(vancomycin)

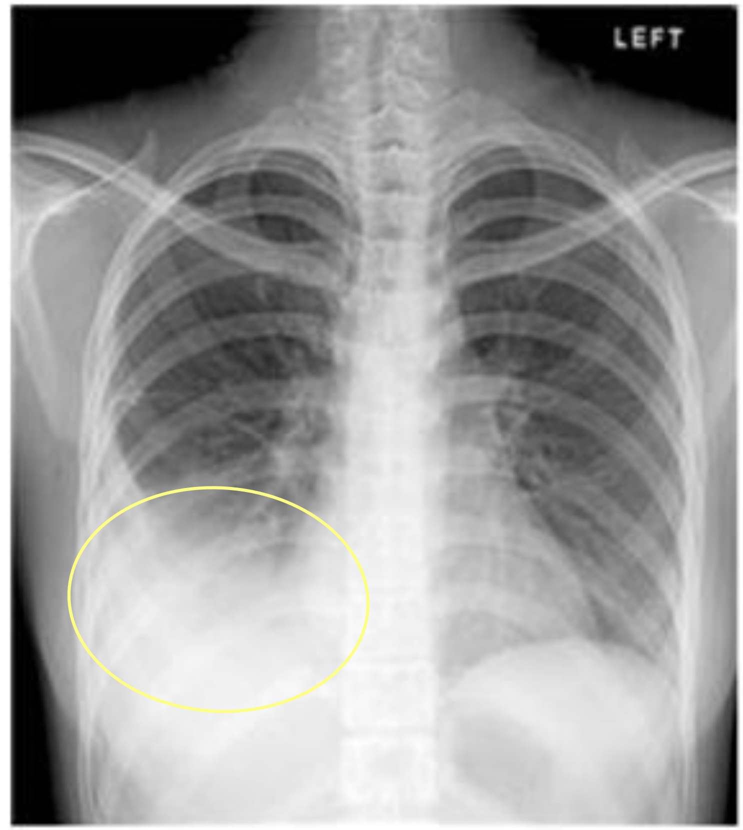
<1st-Exudate vs. Transudate>
Pleural fluid/Serum protein=0.62(>0.5)
Pleural fluid/Serum LDH=2.56(>0.6)
→Exudate
<2nd-원인 감별>
중성구가 81%로 우세+임상증상
→세균성 감염
<가슴관삽입 적응증>
pH6.7(<7.2)
Glucose 25(<60)
→항생제+가슴관 삽입

체중감소→결핵/Cancer

- 73세 남자가 우측 흉부 불쾌감을 주소로 내원하였다. 열, 기침, 가래 등은 없었다. 호흡곤란을 호소하지 않았다. 최근 2개월 사이에 약 3kg 정도의 체중 감소가 있었다. 신체검진에서 혈압은 90/50mmHg, 호흡수 18회/분, 맥박 78회/분, 체온 36.3℃였다. 시진에서 흉부에 외상을 입은 흔적은 없었고 청진에서 우측 폐하부에서 호흡음이 감소되어 있었다. 흉부 방사선 사진에서 흉수가 발견되어 천자를 시행하였고 흉수는 붉은색이며 RBC 15,000/㎕였다. 진단은?

1. 농흉
2. 악성 흉수
3. 결핵성 흉막염
4. 췌장질환과 연관된 흉수
5. 결체조직질환과 연관된 흉수

50세 여자가 2주간의 숨을 쉴 때 아픈 흉통과 호흡곤란을 주소로 내원하였다. 흉부X선 소견은 다음과 같다. 진단을 위해 가장 먼저 해야 할 검사는?



① 가래 그람염색 및 배양검사
② 폐기능검사
③ 기관지경검사
④ 가슴 컴퓨터단층촬영
⑤ 우측으로 누워서 흉부방사선 촬영